Interpretations on the interaction between protein tyrosine Phosphatase and E7 oncoproteins of high and low risk HPV – A computational perception.

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Introduction

Human papillomavirus (HPV) is the most widespread sexually transmitted infection around the globe and it is reported that 75% sexually active women are the victims of the HPV infection. Other factors that is prone for the infection of HPV is the smoking habits, sexual activity at the earliest age and other environmental factors. The cellular hyper proliferation which includes the most prevalent cancer in women is the main association by HPV infection. More than 200 types of HPV has been recognized on the basis of DNA sequence and characterized as the high and low risk genotypes. The high risk types are associated with the cervical, vulvar, vaginal and anal cancers by its interaction with the tumor suppressor proteins whereas the low risk types are associated with genital warts. The HPVs causing cervical cancer in women possess double stranded DNA genomes that are approximately 8 kb in size and encode eight genes among which the E6 and E7 possess transforming properties namely the transmembrane signalling, regulation of the cell cycle, transformation of established cell lines, immortalization of primary cell line and regulation of chromosomal stability. The E6 oncoprotein interacts with the p53 and E7 interacts with pRb which gets degraded and in turn cervical cancer developed. Our study focuses on the identification of interaction of between PTNP14 with the high and low risk types 16 and 11 respectively which helps in understanding the degradation of protein activity.

Objective

The main objective of the study is to identify the difference between the high and low risk interaction with the tumor suppressor protein. Also identifies the major threshold on the residues responsible for cancer and warts

Materials and Methods

1. Retrieval of Transcriptomic data sets from Array express targeting cervical cancer caused by HPV.
2. Analysis of significant genes through network analyst and cytoscape.
3. Gene Ontology is performed with the differentially expressed genes for the molecular function and biological process.
4. Protein – Protein interaction analysis to identify the partners responsible for domination.

Results and Discussion

- HPV 18 – PTPN14 complex is retrieved from the PDB to be used as a reference for the interaction between tumor suppressor and oncoprotein .
- HPV 11 & 16 E7 structure is modeled and checked for conformational stability.
- Protein – protein interaction study is carried.
- Molecular dynamics simulation is carried out for the analysis of the binding conformation stability analysis

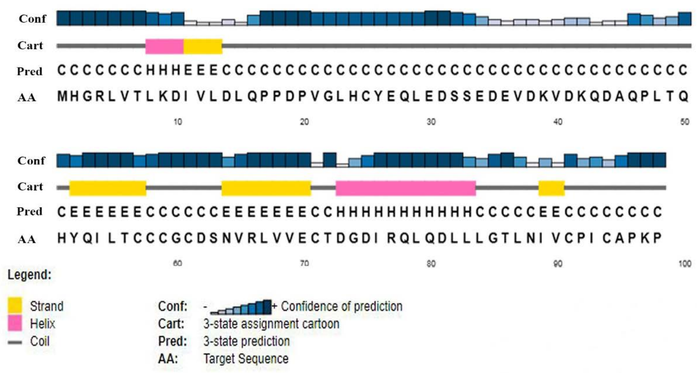

Figure 1: Secondary structure prediction for the E7 oncoprotein of HPV genotype 11

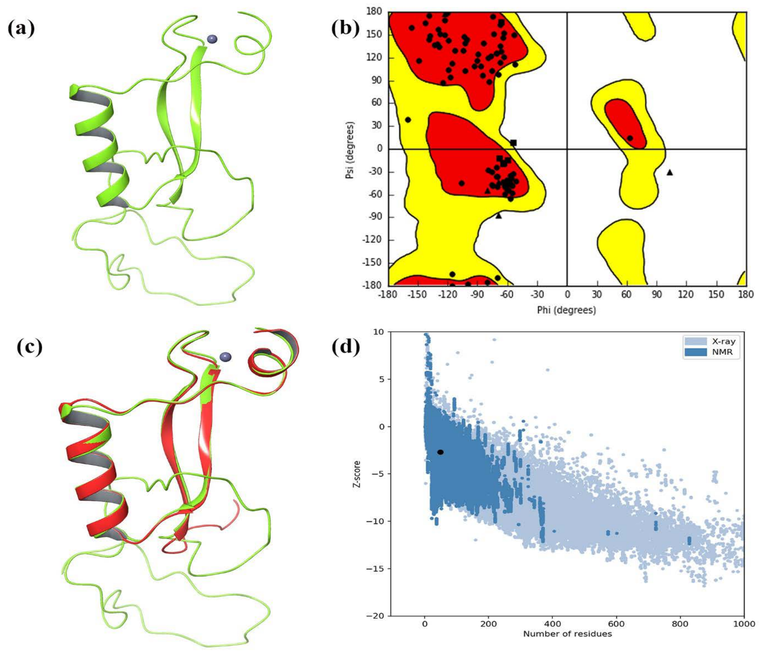

Figure 2: (a) represents the structure of the Modelled HPV 11 E7 and (c) represents the superimposed structure of the template and the modelled structure whereas (b) and (d) represents the Ramachandran plot and the ProSA score as an assessment for the modelled E7 oncoprotein respectively.

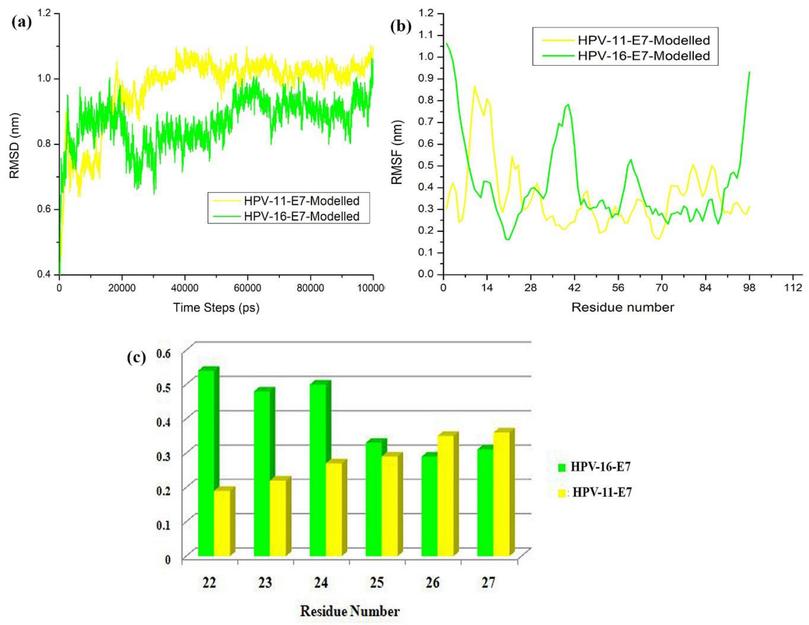

Figure 3: (a) RMSD of the HPV modelled type 11 & 16 E7 (b) RMSF of the HPV modelled type 11 & 16 E7 (c) RMSF mean of the residues present in LXCXE motif.

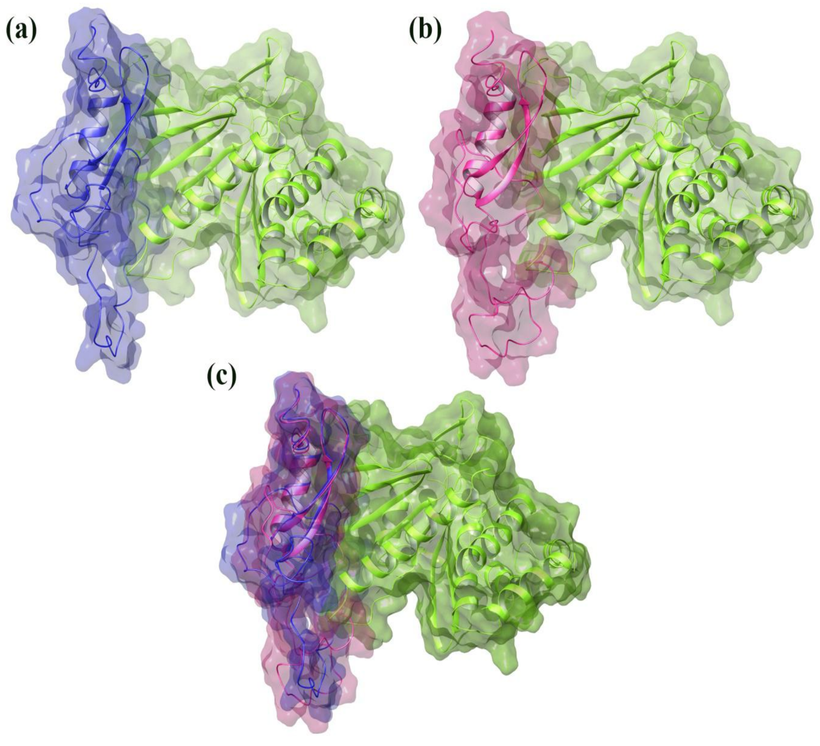

Figure 4: Docked complex of the Protein – Protein complexes (a) HPV 16 E7 – PTPN14 complex, (b) HPV 11 E7 – PTPN14 complex, (c) superimposed structure of the type 16 & 11 complex.

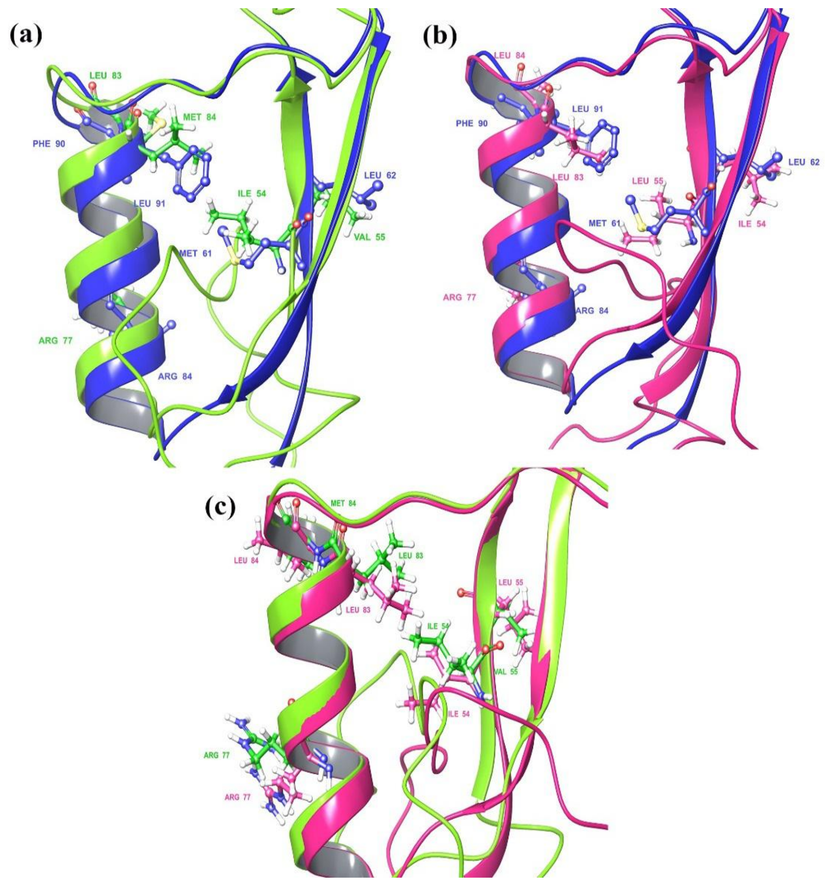

Figure 5: (a) Superimposed structure of the hot spot residues in HPV 18 - 16 E7, (b) HPV 18 - 11 E7 and (c) HPV 11 - 16 E7 respectively

Table 1:Hot spot residues identified with the alignment between the types of E7

S.No	Type of HPV	Residues
1.	HPV 18 E7	Phe 90, Leu 91, Met 61, Leu 62, Arg 84
2.	HPV 16 E7	Leu 83, Met 84, Ile 54, Val 55, Arg 77
3.	HPV 11 E7	Leu 83, Leu 84, Ile 54, Leu 55, Arg 77

Table 2: Interaction scores for the Protein – protein docking

S.No	Complex Description	Cluster Size	Pose Energy	Pose Score
1.	HPV 16 E7 – PTPN14	51	-521.5870	-276.321000
2.	HPV 11 E7 – PTPN14	39	-486.0251	-202.843300

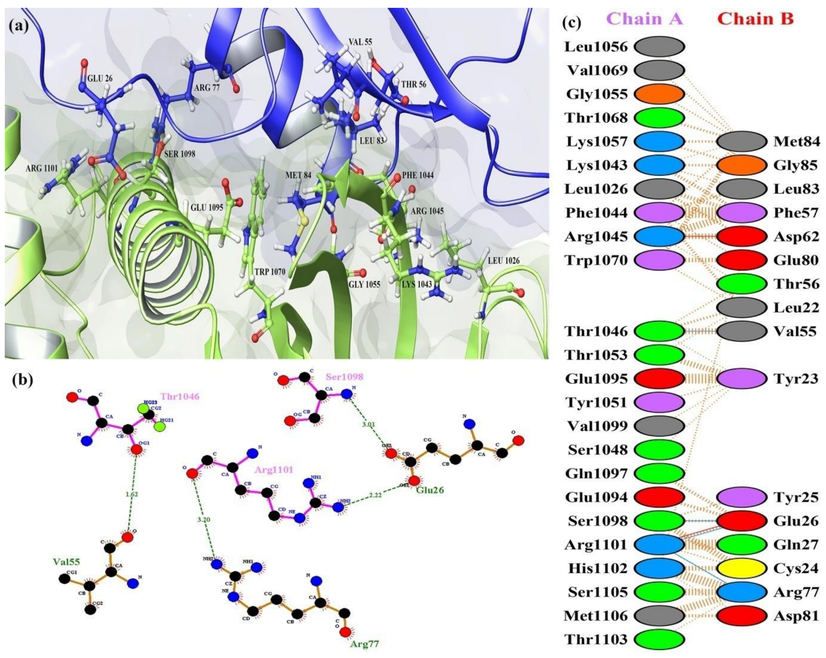

Figure 6: PTPN14- HPV 16 E7 interaction

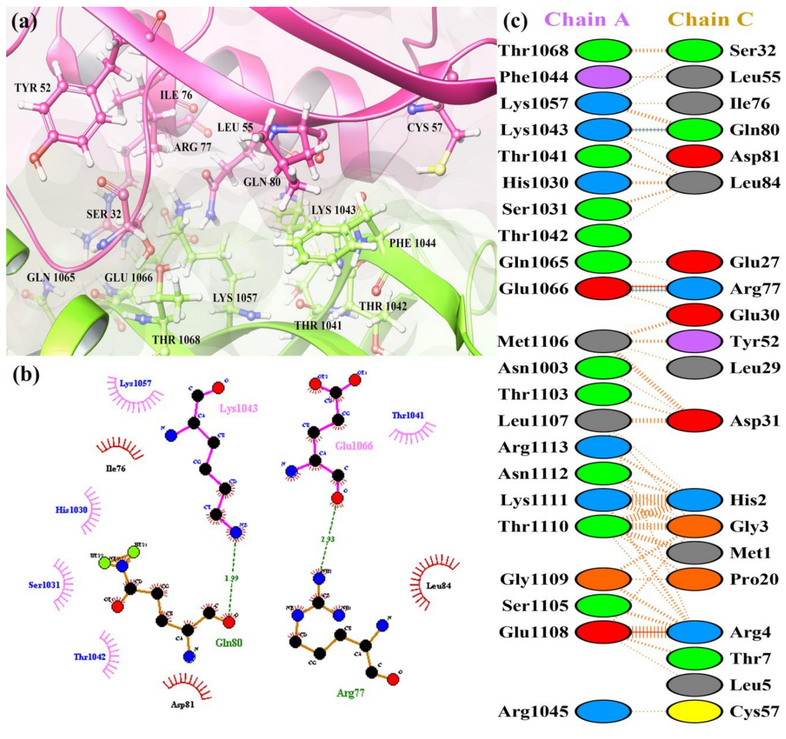

Figure 7: PTPN14 – HPV 11 E7 interaction

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